NPR API Documentation

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❖ Overview

Purpose: NPR validation & submission to backlog API

Scope: Handles NPR intake, sample handling, validation flows

Audience: RFQ

❖ Architecture

High-level system design: Microservices with REST endpoints

Tech: .NET Core 3.1, JSON, SQL

❖ API Endpoints

POST: /NPRWebService/ProcessNPR

Description: Creates NPR request from RFQ

❖ Sample Request

[{"CustomerRfqNo":"R","CustomerCode":"TMMT","OriginatorName":"Singh, Vishwadeep (MTSL)","DeadlineDate":"2026-05-24","ReasonDescription":"TEST2","MaterialNumber":"M15052026_5","MaterialDescription":"Test Mat desc ","CustomerPartNumber":"19062024AC","SupplierPartNumber":"15OCT2024003","BOMUnitOfMeasurement":"BG ","SupplierGroup":"New Supplier","SupplierDetailComments":"Test RFQ","ProjectCode":"177D","QtyPerVehicle":10,"IsSampleDataRequired":true,"SampleSiteDetails":[{"BusinessUnitCode":"AIN1","SampleQty":"10","SendSampleContactPersonName":"Vishwadeep.Singh","RequiredDate":"2026-05-25","FreightAccount":"FA-987654FA-987654FA-987654FA-987654FA-987654FA-90"}],"NPRRaisedRemarks":"Test","ManufacturingSite":"DEVA,KENI","IsCustomerNominatedSupplier":false,"IsCustomerNominatedMaterial":false,"SuggestedManufacturerGroup":"ACAPLAST","AssociatedParts":"NO","NPRUploadSeqNo":"1"}]

❖ Sample Response

{"data":[{"requestID":"SEWS-26-001-009063","error":null,"id":"ea8f1a9d-8723-4509-92dd-a5656df1d322","name":null,"originatorName":"Singh, Vishwadeep (MTSL)","reasonDescription":"TEST2","customerCode":"TMMT","projectCode":"177D","materialDescription":"Test Mat desc ","materialTypeDescription":"Component","manufacturingSite":"DEVA,KENI","customerPartNumber":"19062024AC","supplierPartNumber":"15OCT2024003","materialNumber":"M15052026_5","bomUnitOfMeasurement":"BG","SupplierGroup":"New Supplier","ManufacturerGroup":"New Supplier","supplierDetailComments":"Test RFQ","qtyPerVehicle":"10.0","deadlineDate":"2026-05-24T00:00:00","nprRaisedRemarks":"Test","isCustomerNominatedSupplier":"false","isCustomerNominatedMaterial":"false","isSampleDataRequired":null,"sampleSiteDetails":[{"businessUnitCode":"AIN1","sampleQty":"10","sendSampleContactPersonName":"Vishwadeep.Singh","requiredDate":"2026-05-25","freightAccount":"FA-987654FA-987654FA-987654FA-987654FA-987654FA-90"}],"associatedParts":"NO","suggestedManufacturerGroup":"ACAPLAST","customerRfqNo":"R"}]}

❖ Data Models

Field	Description	Data Type	Length
CustomerRfqNo	RFQ number **(Free Text)**	string	18
CustomerCode	Customer Code **(Master)**	string	18
RequestID	NPR Request ID (Received in response)	string	18
OriginatorName	NPR creator **(Master)**	string	100
DeadlineDate	Date Format yyyy-MM-dd	date	
ReasonDescription	NPR Reason (**Master)**	string	50
MaterialNumber	Material number **(Free Text)**	string	26
MaterialDescription	Description **(Free Text)**	string	15
MaterialTypeDescription	Material Type (Component) **(Master)**	string	50
CustomerPartNumber	CPN **(Free Text)**	string	26
SupplierPartNumber	SPN **(Free Text)**	string	26
BOMUnitOfMeasurement	Unit Of Measure (**Master)**	string	10
SupplierGroup	Supplier Group **(Master)**	string	50
SupplierDetailComments	Required if SupplierGroup has “New Supplier” **(Free Text)**	string	500
ProjectCode	Project **(Master)**	string	18

QtyPerVehicle	Qty	numeric	8,3
IsSampleDataRequired	Flag	bool	true/false
SampleSiteDetails	Required if IsSampleDataRequired is true. Array of sample sites	array	-
NPRRaisedRemarks	Remarks **(Free Text)**	string	500
ManufacturingSite	Site can be comma (,) separated in case of multiple sites **(Master)**	string	max
IsCustomerNominatedSupplier	Flag	bool	true/false
IsCustomerNominatedMaterial	Flag	bool	true/false
SuggestedManufacturerGroup	Supplier Group **(Master)**	string	50
AssociatedParts	YES/NO/UNKNOWN	string	50
NPRUploadSeqNo	Serial number e.g. 1,2,3…	int	
Id	H1 NPR id	guid	36

❖ Data Models (Sample Site Detail)

Field	Description	Data Type	Length
BusinessUnitCode	Business Unit Code **(Master)**	string	4
SampleQty	Sample Qty	int	
SendSampleContactPersonName	Username **(Master)**	string	100
RequiredDate	yyyy-MM-dd	date	
FreightAccount	**(Free Text)**	string	50

❖ Logging & Monitoring

Includes: Application Insights, Telemetry, Tracing, Dependency Tracking, Performance Monitoring.

❖ Validation

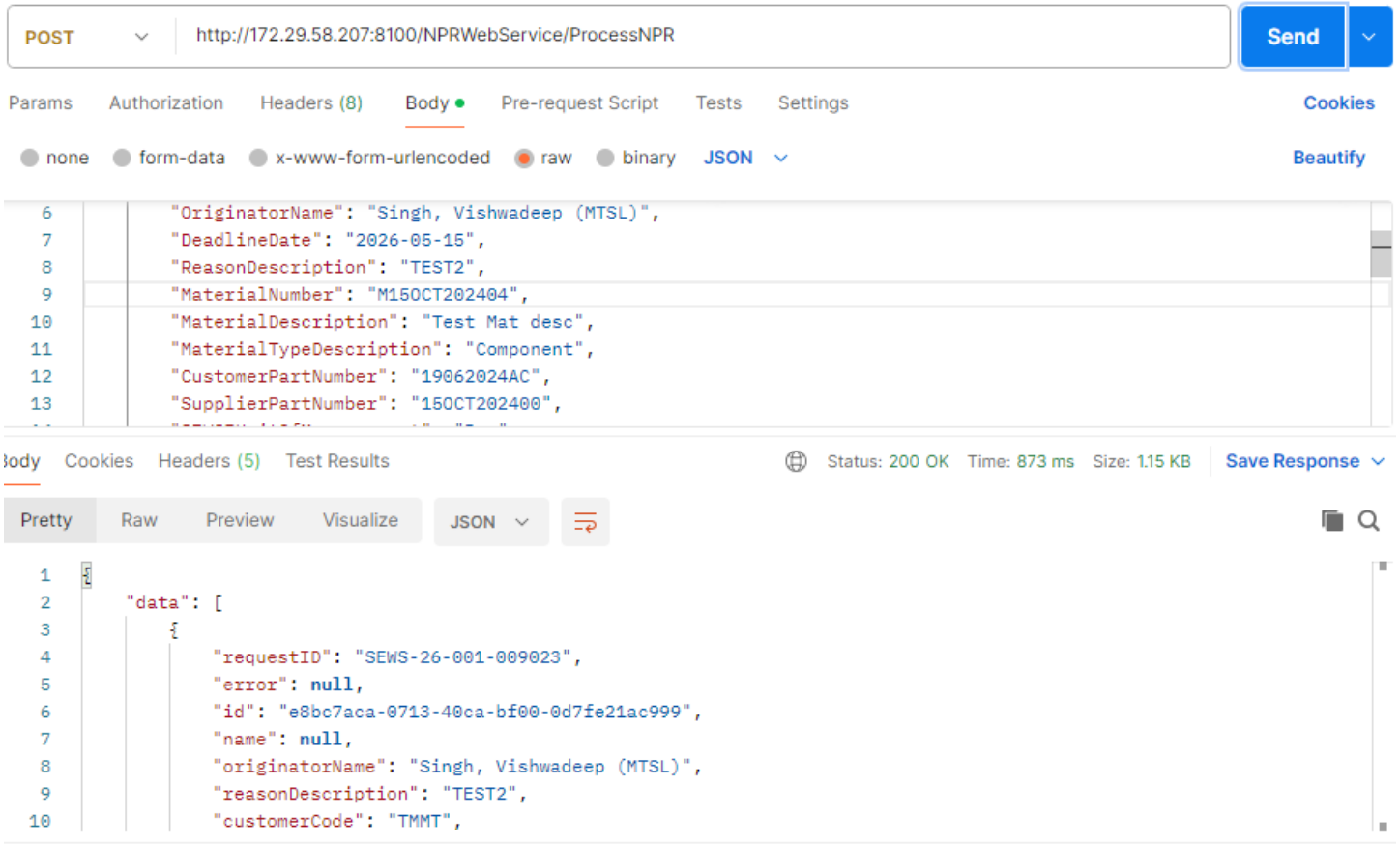
Sr. No.	Field Name	Validation Type	Error Message
1	AssociatedParts	BLANK	Associated Parts: can not be blank
2	AssociatedParts	IncorrectValue	AssociatedParts: Incorrect Value
3	CustomerCode	IncorrectValue	Customer: Incorrect Value
4	CustomerCode	BLANK	Customer can not be blank
5	DeadlineDate	BLANK	DeadlineDate can not be blank, and should be greater than today's date
6	DeadlineDate	LessThanCreationDate	Deadline date can not be less than Creation dates
7	ManufacturerGroup	BLANK	SupplierDetailComments can not be blank when the Supplier Group is "New Supplier"
8	ManufacturerGroup	IncorrectValue	Supplier Group: Incorrect Value
9	ManufacturingSite	IncorrectValue	Manufacturing Site: Incorrect Value
10	Material Description	LENGTH	Material Description Can not be less than 1 and greater than 15 characters.
11	MaterialDescription	BLANK	MaterialDescription Can not be blank!
12	MaterialNumber	LENGTH	Sumitomo Number Can not be less than 8 and greater than 26 characters.
13	MaterialNumber	DUPLICATE	[50464] Duplicate Materials exist, please correct.
14	MaterialNumber	BLANK	MaterialNumber Can not be blank!
15	MaterialNumber	EXIST	Part Number already exists in H1
16	MaterialNumber	EXIST	Part Number already exists for another NPR @RequestId
17	MaterialNumber	LENGTH	MaterialNumber Can not be blank and Can not be less than 8 and greater than 26 characters
18	MaterialTypeDescription	BLANK	MaterialTypeDescription can not be blank
19	MaterialTypeDescription	IncorrectValue	Material Type Description: Incorrect Value
24	ProjectCode	BLANK	ProjectCode can not be blank
25	ProjectCode	IncorrectValue	Project Code: Incorrect Value
26	ProjectCode	BLANK	ProjectDetails Can not be blank!
27	QtyPerVehicle	Grater than zero	QtyPerVehicle Cannot be Zero or less than Zero
28	ReasonDescription	BLANK	ReasonDescription can not be blank
29	ReasonDescription	IncorrectValue	Reason Description: Incorrect Value
30	ReasonDescription, CustomerCode, ProjectCode	DUPLICATE	Combination of fields ReasonDescription, CustomerCode, ProjectCode must be same
31	SampleSiteDetails	BLANK	If Sample Required is True, provide Sample Site, Sample Quantity (greater than 0),

			Sample Recipient, and a Sample Required Date that is today or later.
32	SampleSiteDetails.BusinessUnitCode	IncorrectValue	Sample Site:: Incorrect Value
33	SampleSiteDetails.SampleQty	Incorrect SampleQty	Sample Quantity cannot be greater than commodity sample quantity
34	SendSampleContactPersonName	IncorrectValue	Sample Recepient: Incorrect Value
35	BOMUnitOfMeasurement	IncorrectValue	UOM: Incorrect Value
36	SuggestedManufacturerGroup	IncorrectValue	Suggested Manufacturer: Incorrect Value
37	SuggestedManufacturerGroup	BLANK	Suggested Manufacturer Can not be blank!
38	SupplierDetailComment	BLANK	SupplierDetail Can not be blank!
39	SupplierPartNumber	BLANK	Supplier Part number cannot be blank
40	SupplierPartNumber	DUPLICATE	Duplicate Supplier Part Number exists, please correct.
41	SupplierPartNumber	BLANK	Supplier Part Number already exists for another NPR @RequestId
42	SupplierPartNumber	BLANK	Provide either the Material Number or the Supplier Part Number.
43	SupplierPartNumber, MaterialNumber	BLANK	Both SupplierPartNumber and Sumitomo Number can not be blank
44	VehicleTotalQuantity	Grater than zero	Total Quantity Cannot be Zero or less than Zero

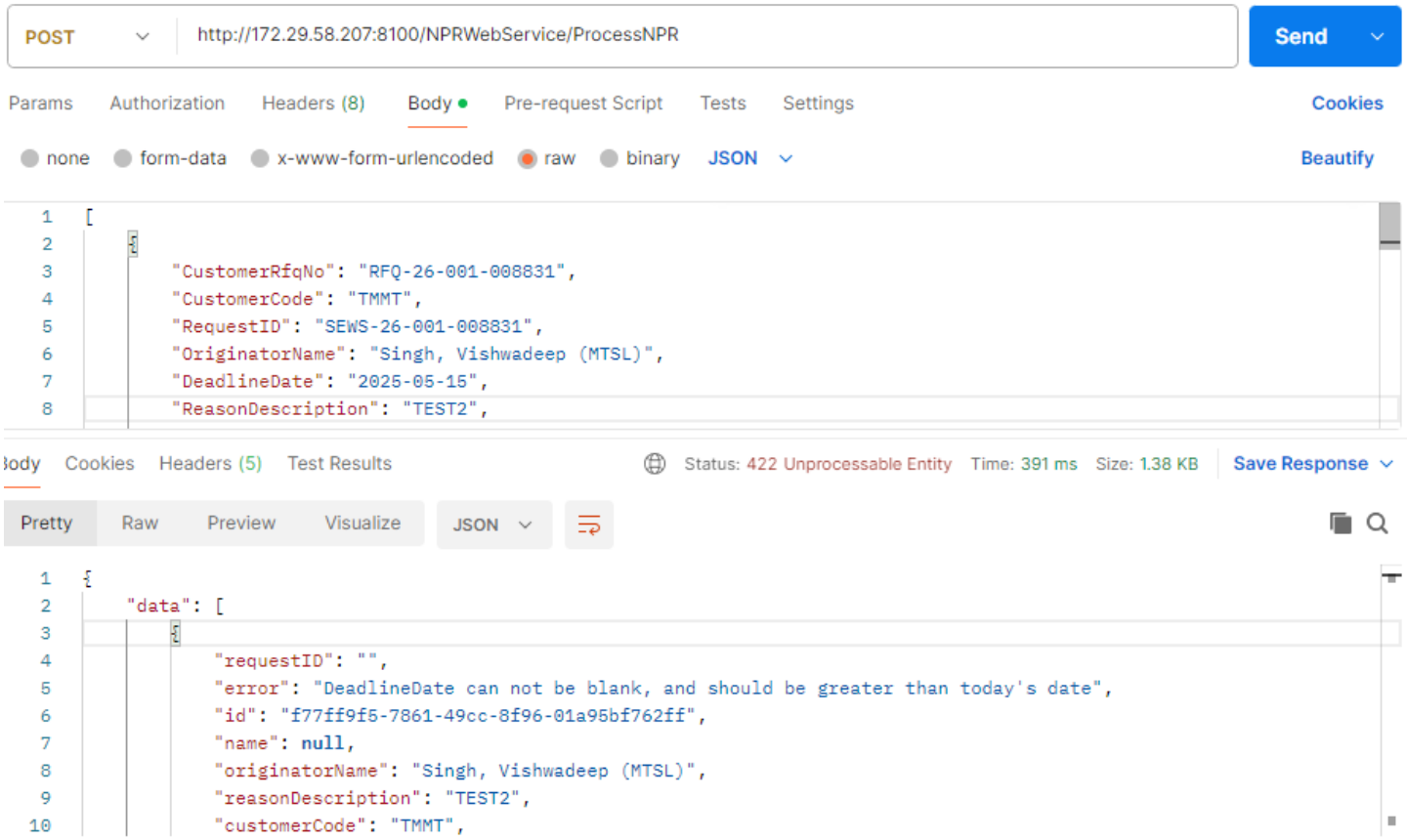
❖ Appendix

➢ Calling from Postman:

1. Validation Success.



2. Validation failed.



➢ HTTP Status Code:

Status Code	Description
422	In case of validation Failed.
200	Successfully process all NPR.
500	API break (Internal server error).

➢ Mater Field Description:

Json Field Name	Table Name	Column Name
CustomerCode	[H1.GlobalMasters].dbo.CustomerHierarchy	CustomerCode
OriginatorName	[H1.Core].dbo.UserList	Logon
SupplierGroup	[H1.GlobalMasters].dbo.SupplierGroup	SupplierGroupName
ManufacturingSite	[H1.GlobalMasters].dbo.BusinessUnit	LegacyBusinessUnitCode
MaterialTypeDescription	Only accept value "Component"	
ProjectCode	[H1.GlobalMasters].dbo.PlatformHierarchy [H1.GlobalMasters].dbo.ProgramHierarchy [H1.GlobalMasters].dbo.ProjectHierarchy [H1.GlobalMasters].dbo.ProjectEngineeringAssociation [H1.GlobalMasters].dbo.CustomerHierarchy [H1.Core].dbo.UserList	ProjectName
ReasonDescription	[H1.Purchasing].dbo.RFQReason	ReasonDescription
BusinessUnitCode	[H1.GlobalMasters].dbo.BusinessUnit	LegacyBusinessUnitCode
SendSampleContactPersonName	[H1.Core].dbo.H1UserRegister	ADLogon
BOMUnitOfMeasurement	[H1.GlobalMasters].dbo.UnitOfMeasureAssociation [H1.GlobalMasters].dbo.UnitOfMeasure	SourceUnitOfMeasureCode
SuggestedManufacturerGroup	[H1.GlobalMasters].dbo.SupplierGroup	SupplierGroupName